



Diabetes challenges and promising solutions based on the internet of medical things, blockchain, and artificial intelligence: A systematic literature review

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Abstract

Diabetes Mellitus affects 8.8% worldwide and is projected to reach 10.9% by 2045, increasing the need for advanced diagnosis and treatment. Technology-driven solutions are recommended for better management and timely interventions. Recently, the Internet of Things (IoT) has penetrated the medical sector, presenting the Internet of Medical Things (IoMT) as a new paradigm. Moreover, it has been integrated with technologies such as blockchain to enhance security levels and Artificial Intelligence (AI) to improve decision-making. This paper presents a concentrated study of the most relevant work that links IoMT with industry to help researchers, developers, and medical service providers accelerate improvements in medical services. Diabetes is selected as a case study. This paper reviews the most recent work on remote patient monitoring (RPM) and continuous glucose monitoring (CGM) to improve decision support systems, increasing the precision of prediction of critical events related to diabetes in daily life. In addition, it highlights the role of blockchain-based platforms in improving security, integrity, immutability, and decentralized healthcare solutions. Given the increasing focus on industrial diabetes management solutions, a comparative analysis of existing IoMT-based diabetes therapy systems is presented, shedding light on promising technologies that could fulfill essential requirements to enable smarter and more secure healthcare ecosystems. Given the urgency of this global challenge, diabetes management has become one of the most attractive research directions, especially when combined with emerging technologies such as IoMT, blockchain, and AI, which open new frontiers for smarter and more secure healthcare. The review was conducted using major and trusted digital libraries, including IEEE Xplore, SpringerLink, ScienceDirect, PubMed, and SAGE Journals. Coverage was cross-verified using Scopus and Web of Science to ensure completeness. In total, 1,120 papers were retrieved, of which 127 met the inclusion criteria for detailed analysis.

Keywords Diabetes · Insulin Pump Therapy · Artificial Continuous Glucose Monitoring (CGM) · Remote Patient Monitoring (RPM) · Internet of Medical Things (IoMT) · Blockchain · Artificial intelligence (AI)

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1 Introduction

Diabetes Mellitus (DM), more often known as simply "Diabetes", is a chronic metabolic disease that affects 8.8% of the global population. The number of diabetics is predicted to reach 10.9% (783 million people) by 2045 [1–4]. It is considered one of the top rapidly proliferating chronic diseases. [5]. DM is characterized by abnormal blood sugar oscillations. There are two critical events that thread diabetics all the time; (a) high blood sugar, also referred to as hyperglycemia, occurs when there may be too much glucose (sugar) circulating inside the bloodstream, and (b) low blood sugar, additionally referred to as hypoglycemia, happens when

blood glucose levels drop below the normal variety (usually underneath 70 mg/dL), which may cause coma [4, 6]. Thus, occurring such situations can lead to various health challenges if not managed well in the time [7]. Challenges and illnesses that DM is related to and may cause, include issues with the kidneys, blood vessels, heart diseases, and some types of blindness [7–9].

In the last decade, Healthcare 4.0 [10, 11] integrates multiple technologies such as cloud computing and Internet of Things (IoT) to present telemedicine applications. Recently, due to powerful penetration of the IoT into the medical sector, it is known as the Internet of Medical Things (IoMT) [12]. In general, the IoT plays a critical role in our life, where wearable devices and mHealth have been joined variety of applications [5, 13, 14]. IoT devices are known as smart sensors, smart objects or Smart Things (SThs) [15, 16], they can sense and monitor certain feature in real-time (e.g., glucose level). Wearable and IoT systems that enable continuous glucose monitoring (CGM) are promising solutions for diabetes. Recent studies emphasize that IoMT ecosystems face serious risks of data breaches, making blockchain a vital technology to ensure confidentiality, integrity, and regulatory compliance [17]. Further analyses have shown that while blockchain integration strengthens privacy and interoperability, challenges of scalability and standardization remain central to secure IoMT deployment [18].

The increasing number of diabetes patients [19] and their continuous need for follow-up to adjust medication doses require modern methods for quick communication and saving time for doctors [20], whose numbers are no longer proportional to the growing number of patients. As a result, IoT-based systems are recommended for enabling sensible remote patient monitoring (RPM). RPM offers benefits such as enable more accurate patient monitoring with healthcare convenience, provide real-time connecting with healthcare providers, reduce the cost of medical services, and improve the healthcare quality [21, 22]. For diabetes, IoT-based systems should meet set of special issues such as data integrity, security and privacy. In the contemporary landscape of

healthcare informatics, the imperative to fortify data security and privacy has drawn parallels with principles traditionally associated with electronic warfare. As the specter of cyber-attacks on critical healthcare infrastructure looms large, the need for robust security measures has become increasingly urgent [23, 24]. Blockchain plays a critical role in healthcare system security [14, 25]. AI and machine learning (ML) algorithms are being used to analyze huge datasets of glucose readings, insulin dosages, and other patient-related information. These algorithms can be used to identify patterns, predict blood sugar fluctuations, and provide recommendations, improve decision support systems [26]. To address performance and scalability, hybrid frameworks combining blockchain with edge computing have been introduced, enabling lightweight authentication with efficiency above 98% [27]. Similarly, federated learning combined with blockchain provides privacy-preserving collaboration across IoMT devices while achieving high accuracy in intrusion detection and disease prediction [28]. Beyond technical designs, trust and dependability are increasingly recognized as critical for patient adoption of AI–blockchain MedIoT applications, where reliability and transparency directly impact healthcare outcomes [29]. Finally, the emergence of generative AI integrated with IoMT and blockchain introduces advanced possibilities for predictive analytics, personalized therapies, and secure management of electronic health records, pointing toward a new paradigm of intelligent and secure healthcare [30].

To provide a conceptual overview, Figure 1 illustrates how diabetes challenges can be addressed through IoMT devices, supported by blockchain for data security and AI/ML for predictive analytics, ultimately leading to promising healthcare solutions.

1.1 Motivation

Recently, number of diabetics has significantly increased worldwide, which attracts researchers to spot their light on this theme [19, 31]. According to Wisenvoter [2], the global average rate for diabetes till 2024 is about 8.8%. Fig. 2 shows top-10 countries with highest diabetes rates in the world. Consequently, researchers spot their light on this theme to find solutions to increase accuracy of diabetes diagnoses and to improve treatment approaches. In particular, integrating IoMT with blockchain and AI is increasingly spotlighted as a promising and attractive research theme, not only because of the medical need but also due to the technological opportunities it creates for innovation. This convergence offers researchers fertile ground to explore secure, intelligent, and scalable healthcare systems, making the topic highly appealing for both academic and industrial communities. Fig. 3 shows search analysis report for

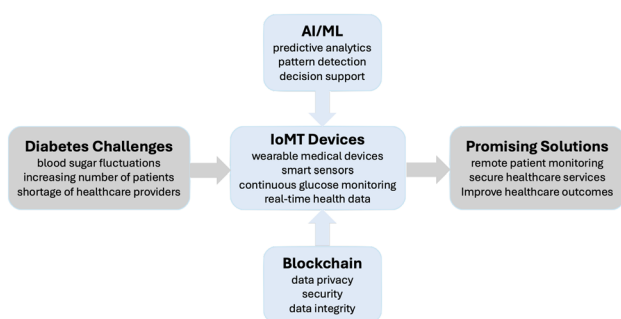


Fig. 1 Conceptual overview linking diabetes challenges with IoMT-based monitoring, blockchain-enabled security, and AI-driven analytics to achieve promising healthcare solutions

1	Pakistan	30.80%
2	Kuwait	24.90%
3	Northern Mariana Islands	23.40%
4	New Caledonia	23.40%
5	Nauru	23.40%
6	Marshall Islands	23%
7	Mauritius	22.60%
8	Egypt	20.90%
9	Solomon Islands	19.80%
10	Qatar	19.50%

Fig. 2 Wisevoter report [2]: The top-10 countries with the highest diabetes rates in the world

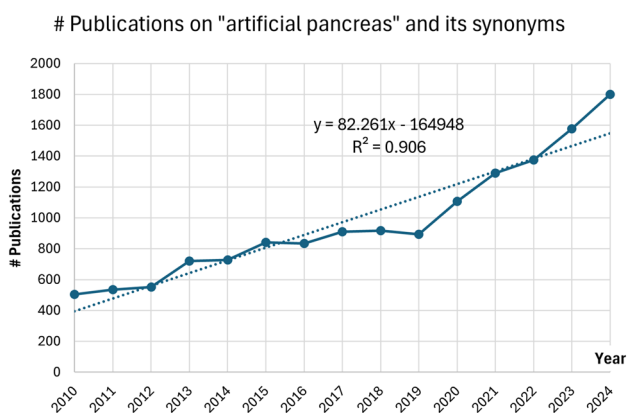


Fig. 3 Scopus [32]: Search Results from 2010 to 2024 using a query on "Artificial Pancreas" and its synonyms

Table 1 Eligibility criteria

Inclusion	Exclusion
Peer-reviewed Article/Review/Conference (2011–2025), English.	Non-English; pre-2011; abstracts/ editorials/ posters; inaccessible full text.
Diabetes-focused <i>IoT/IoMT</i> in monitoring/therapy (CGM/RPM/insulin/telemedicine/wearables).	Generic <i>IoT/AI/Blockchain</i> without diabetes- <i>IoMT</i> relevance.
Integrates <i>AI/ML</i> and/or <i>Blockchain/Security</i> , or provides implementational/empirical insights.	Insufficient methodological detail; duplicates.

a query on diabetes and IoT using Scopus database [32], spanning 2010 to 2024. This chart reflects how researches are interested on this theme.

1.2 Contributions of This Survey

Main contributions of this paper are summarized in the following points:

- Presenting a comprehensive background on diabetes.
- Providing an overview for the IoMT, Blockchain, and AI technologies, discussing their solutions for diabetes.
- Studying synergies (challenges and integration opportunities) between IoMT, Blockchain, and AI technologies.
- Surveying industrial IoMT systems for diabetes (artificial pancreas).
- Highlighting future directions and recommendations for improving diabetes management systems.

1.3 Organization of This Survey

The remainder of this paper is organized as follows. Section 4 presents a background on diabetes clarifying main key issues concerning diagnosis and treatment. Section 5 presents an overview on the IoMT, as it is the under layer technology; where ICT integration occurs. Section 6 discusses potential power of integrating blockchain and AI in IoMT. Section 7 presents a comparable study for industrial IoMT for diabetes (smart pancreas). Section 8 recommends some features for enabling secure smart healthcare systems to put researchers on the right path catching main pillars for aiding diabetes and improving healthcare sector in general. Section 9 presents a conclusion.

2 Methodology

We conducted a Systematic Literature Review (SLR) with a clear, reproducible protocol: research questions (RQs), data sources, concept-based search strategy, staged screening with counts, inclusion/exclusion (I/E) criteria, quality assessment (QA), year-wise final selection, and a PRISMA-style flow diagram.

2.1 Research Questions

- RQ1: What are the main Internet of Medical Things (IoMT) applications for diabetes management (e.g., CGM, RPM, insulin pumps/closed loop) and how are they realized?
- RQ2: Which security, privacy, and interoperability challenges arise in diabetes-focused IoMT ecosystems?
- RQ3: How do *blockchain* and *AI/ML* integrate with IoMT to strengthen security/privacy and enhance prediction/decision support?

RQ4: What gaps and future directions emerge for robust, secure, and intelligent IoMT-based diabetes care?

2.2 Data Sources

We queried major scholarly indices (Scopus and Web of Science) and retrieved full texts from leading digital libraries (IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, PubMed, SAGE Journals), adapting fielded Boolean syntax per database.

2.3 Search Strategy

Concept groups and their synonyms were combined in advanced, fielded searches:

- **Diabetes** (Type 1/2, glycemic control, hypo/hyperglycemia);
 - **IoT/IoMT** (Internet of Things/Medical Things, wearables, telemedicine/mHealth);
 - **Monitoring/Therapy** (CGM, RPM, insulin pumps, artificial pancreas, glucose sensors);
 - **AI/ML** (machine/deep learning, prediction, decision support, federated learning);
 - **Blockchain/Security** (blockchain, distributed ledger, smart contracts, privacy, access control, encryption).
- Filters:** Years 2011–2025; document types: Article, Review, Conference Paper; language: English; subject areas: Medicine, Computer Science, Engineering.

2.4 Screening Pipeline and Counts

Identification. Baseline diabetes+IoT/IoMT search returned **1,120** records; applying the standard facets yielded **768**.

Use-case focus. Restricting to IoMT monitoring/therapy terms (CGM/RPM/insulin/telemedicine) produced **406**; with facets re-applied for that run, **310** remained.

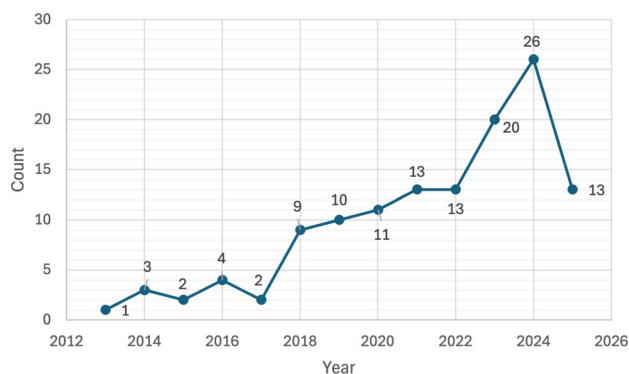


Fig. 4 Year-wise distribution of included studies

Technology subsets. Two focused sets were obtained: **AI/ML (150)** and **Blockchain/Security (28)**. Their exports were merged and deduplicated by Title/DOI to form the eligibility pool.

Final inclusion. After full-text assessment using the I/E and QA below, the synthesis included $N_{\text{final}} = 127$ studies (year-wise distribution in Fig. 4).

Index coverage and auditability. Identification counts are reported strictly from Scopus/WoS at the search date. Where a relevant peer-reviewed study was located outside those indices (e.g., publisher portals or preprint servers), it was screened under the same I/E and QA rules, deduplicated, and included in N_{final} only if fully compliant. A time-stamped registry of exports, DOIs, and de-duplication logs is archived and available upon request.

2.5 Inclusion and Exclusion Criteria

2.6 Quality Assessment (QA)

Candidates were appraised on three binary criteria (1/0); studies satisfying *C1* & (*C2* or *C3*) progressed.

2.7 Final Selection and Year-wise Distribution

Years were taken *exactly as listed in the final References*. To improve readability in narrow columns, the distribution is shown vertically:

2.8 PRISMA Flow

The staged identification–screening–eligibility–inclusion procedure is summarized in Fig. 5: starting from 1,120 records, standard filters yielded 768; adding IoMT monitoring/therapy terms produced 406 (310 after re-applying facets); technology-focused subsets returned AI=150 and Blockchain/Security=28; after merging, de-duplication, and full-text checks, we included $N_{\text{final}} = 127$ studies.

3 Literature review

Different databases such as Scopus and Web of Science are queried and references are filtered based on main objectives, goal, and intended contributions for this paper. Table 3 presents a brief summary of recent top-10 related work published in prestigious journals since 2022, highlighting their main contributions. To sum up, none of them neither present diabetes case study nor present comparative study for industrial IoMT.

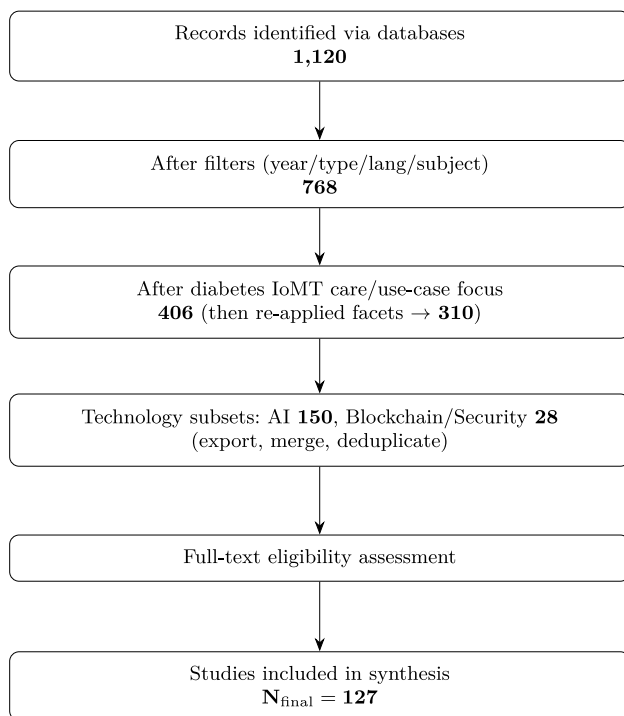


Fig. 5 PRISMA-style flow of study selection

Beyond theoretical frameworks, several practical examples demonstrate that IoMT, blockchain, and AI technologies are being applied in real healthcare contexts. In [33], diabetes management practices in India and the United States were analyzed, showing how AI-based systems support personalized treatment recommendations, remote patient monitoring, and automated insulin delivery. The study emphasizes that large populations and limited resources necessitate such advanced digital interventions. In [34], a lightweight blockchain and fog-enabled remote patient monitoring system was proposed and evaluated, with simulation results showing a 40% improvement in responsiveness compared to traditional cloud-only models, proving feasibility in real-time chronic disease monitoring. Furthermore, [35] introduced an AI-driven IoMT telemedicine framework that integrates IoMT, cloud computing, and deep learning for disease classification, achieving 98.62% accuracy in experimental evaluation. These examples confirm that the integration of IoMT, blockchain, and AI is not confined to conceptual models but is being increasingly tested and implemented in diverse healthcare systems.

A critical limitation of much of the current research is the lack of empirical validation in real-world healthcare environments. As summarized in Table 3, many recent frameworks remain at the conceptual or simulation stage, with clinical testing and large-scale deployments still pending. This highlights the importance of moving beyond theoretical proposals toward practical trials and real-world

evaluation, which are essential to ensure the safety, reliability, and effectiveness of IoMT, AI, and blockchain-based solutions in diabetes management. Examples of industrial deployments, such as the FDA-approved hybrid closed-loop insulin delivery systems summarized in Table 8, demonstrate that such validation is feasible and provide valuable models for future research translation.

As summarized in Table 3, existing works reveal persistent gaps in predictive analytics, data privacy, interoperability, and trust within IoMT-based diabetes management. Artificial Intelligence (AI) has been proposed to bridge these gaps through predictive modeling, personalized decision support, and proactive monitoring, while blockchain offers decentralized trust, privacy-preserving data exchange, and standardized interoperability across heterogeneous systems. These synergies are further discussed in 6, where we outline in more detail how AI and blockchain technologies can complement IoMT to address the identified limitations and enhance diabetes management.

4 Diabetes mellitus(DM)

Diabetes Mellitus (DM) is a common global endocrine and chronic metabolic disease characterized by persistent increased blood glucose levels (hyperglycemia). This disease can be caused by either defects in insulin secretion or resistance to its insulin action [46]. Beta islets of pancreas secrete insulin hormone which regulates blood levels of glucose. Diabetes is manifested by multiple clinical manifestations and can cause a wide range of complications including nephropathy, neuropathy, vasculopathy and comas [47]. DM has a significant impact on patient's quality of life, morbidity and economic status [48]. In 2021, there were 537 million diabetic patient worldwide and it is expected that this number would increase to reach up to 7 hundred million patient by 2045 [49]. Early detection and proper glycemic control helps much to control and reduce diabetes complications and its burden [48].

4.1 Types of DM

For more precise, there are many subtypes of DM including maturity onset diabetes of the young diabetes, due to diseases of the exocrine pancreas and drug induced DM [48]. The most common three types of DM are as follows.

- **Type-1 DM** is an autoimmune disorder characterized by formation of autoantibodies against beta cells of pancreas leading to their destruction and depletion. This in turn leads to decrease in insulin serum level and subsequent hyperglycemia. This subtype occurs more frequently in

Table 3 A Summary for Recent Related Work

Ref.	Year	Main Topics	Main Contributions	Research Gaps
[30]	2025	Generative AI + IoT + blockchain in healthcare. Applications, ethics, and interoperability.	Comprehensive synthesis of GenAI use (predictive analytics, EHR, assistants). Links blockchain to trustworthy data pipelines. Recommends scalable designs and standardized data exchange.	Open issues in scalability/energy for blockchain. Governance and ethical frameworks for GenAI remain under-specified.
[27]	2025	IoMT authentication at scale. Hyperledger Indy + edge computing. Lightweight crypto (e.g., proxy re-encryption).	Hybrid, permissioned framework for low-latency auth. Consortium governance; APBFT-style consensus. Reported gains: ~2.93% lower latency, 98.33% auth efficiency.	Needs evaluation across heterogeneous devices and networks. Energy/performance trade-offs and portability to non-Indy stacks.
[28]	2025	Federated learning + blockchain for IoMT. Intrusion detection and proactive monitoring.	FBCI-SHS framework combining FL, BC, IoMT. Reported metrics: privacy/security 98.73%, IDS 97.16%, disease detection 96.43%, proactive mgmt 98.37%, interoperability 96.74%.	Clinical validation and regulatory adoption pending. Robustness under non-IID data and cross-institution drift not fully explored.
[29]	2025	Trust and dependability in AI–blockchain MedIoT. Reliability, privacy, ethics, transparency.	Identifies key trust challenges and future directions. Frames patient/practitioner perspectives for dependable MedIoT. Guidance for trustworthy AI + blockchain integration.	Lack of quantitative trust metrics and evaluation protocols. Explainability and certification pathways remain open questions.
[36]	2024	Integration of AI and IoT. Security and Privacy in IIoT. Challenges and Future Directions.	Comprehensive Survey on IIoT. Taxonomy and Lessons Learned. Proposals for Security and Privacy Countermeasures. Future Research Insights.	Lack of detailed implementation strategies for security and privacy solutions.
[37]	2024	IoMT Security. Role of AI in IoMT Security. Cybersecurity Threats in IoMT. AI-driven Solutions in IoMT.	Systematic Review of IoMT Security. Highlighting AI's Role in Enhancing Security. Proposal for AI-enhanced Security Frameworks. Research Gaps Identification.	Limited real-world case studies for validating the proposed security frameworks.
[38]	2024	IoMT Overview. IoMT Integration with AI. IoMT Integration with Blockchain.	Highlighting IoMT Architecture. Identification of Challenges in IoMT. Emphasis on Data Security, Privacy, and Advanced IoMT Technologies. Future Trends in IoMT.	Lack of empirical analysis on blockchain impact in IoMT.
[39]	2024	IoMT Applications. Data Fusion in IoMT. Security Challenges and Role of Blockchain in IoMT.	Identification of Research Gaps. Proposing a Framework for IoMT Security. Enhancing IoMT Data Analysis with AI. Future Recommendations for IoMT.	Security framework is conceptual; lacks implementation and testing details.
[40]	2024	IoMT Overview: concepts, applications, challenges, and adaptation solutions. AI impact on IoMT.	Comprehensive Overview on IoMT. AI Applications in IoMT. Security Issues and Methodologies. Challenges and Future Opportunities.	Scalability issues of AI adoption in IoMT were not provided.
[41]	2023	IoT for Diabetes. Cybersecurity Risks and Adoption Issues.	Analysis of Data Privacy and Security Concerns. Clarify Technological Complexity. Discuss Challenges of Current Healthcare Systems. Presents Recommendations to Improve IoT Adoption.	Technical solutions for cybersecurity threats were not provided.
[42]	2022	IoMT Overview: (Applications, Sensors and Wireless Communication Technologies). Machine Learning (ML), Cloud Computing and Edge Computing in IoMT.	Review of Sensors and Technologies. Discussion on Communication Protocols. Exploration of ML Techniques. Highlighting Challenges. Future Directions in IoMT.	Lacks discussion on regulatory and ethical concerns of IoMT.
[43]	2022	IoMT Overview: (Industrial Applications, Data Acquisition and Layer Management). ML and Cloud Computing in IoMT.	Comprehensive Review of IoMT Technologies. Emphasis on Real-Time Applications. Identifying Challenges in IoMT. Integration with ML. Proposals for Future Development.	Empirical validation of proposed models are limited.

Table 3 (continued)

Ref.	Year	Main Topics	Main Contributions	Research Gaps
[44]	2022	IoMT Overview: (Applications and Challenges). ML in IoMT. Blockchain Integration in IoMT.	Review of IoMT Architecture and Applications. Exploration of Emerging Technologies in IoMT. Proposals for Solving IoMT Challenges. Analysis of Complementary Technologies. Future Directions for IoMT.	Interoperability challenges in IoMT architectures were not provided.
[45]	2022	IoMT Overview: (Architecture IoMT and Emerging Technologies). Case Studies.	Introduction to Cloud-Fog in IoMT Architecture. Review of Emerging IoMT Technologies. Experimental Case Studies. Recommendations for Future Research.	Limited real-world implementation examples of proposed technologies.

Table 4 Major differences between Type-1 DM and Type-2 DM [52]

Parameter	Type-1 DM	Type-2 DM
Age of onset	usually < 10 years	usually in adults
Family history	positive only in 10% of cases	positive in > 70% of cases
Body weight	normal/under weight	usually Obese
Insulin resistance manifestations	absent	present
Pancreatic auto antibodies	present	absent
Ketoacidosis at 1 st presentation	more common	less common
Insulin & c-peptide	low or absent	variable
Insulin dependency	Insulin dependent	variable

children and young adults. This subtype patients are in a need for insulin therapy for life [49]. Recent updates indicate that patients with Type 1 DM continue to face significant challenges such as glycemic variability, hypoglycemia, and long-term complications. Advances in IoMT technologies, particularly continuous glucose monitoring (CGM) and closed-loop insulin delivery systems, have demonstrated better prediction of glucose fluctuations and improved metabolic control [49, 50]. These tools enable proactive management and reduce risks during sensitive periods such as pregnancy, where maintaining tight glycemic control is critical [50].

- **Type-2 DM** is the most common subtype and characterized by body tissues resistance to the circulating insulin leading to hyperglycemia and other metabolic effects [51]. Obesity, unhealthy life style and lack of physical activity are its major risk factors. In Type 2 DM, recent evidence highlights unmet needs during pregnancy and in younger populations. Women with T2DM in pregnancy often present with suboptimal glycemic control, increasing the risk of preeclampsia, neonatal complications, and cesarean delivery [50, 51]. Furthermore, in children and adolescents, T2DM is increasingly recognized as an aggressive condition with rapid β -cell decline and early development of complications [52]. IoMT-enabled lifestyle monitoring, remote consultations, and decision-support platforms offer new

strategies to address these gaps by promoting adherence and early intervention [48, 49].

- **Gestational DM** is a subtype occurs during pregnancy and may disappear after delivery [53]. The women who had gestational DM are at increased risk of type-2 DM development later on her life [50]. Furthermore, gestational DM increases the risk of fetal complications such as macrosomia, preterm labour, neonatal hypoglycemia or even stillbirth. It also increases maternal complications such as hypertension, pre-eclampsia and infections. Updated data show that women affected are at substantially higher long-term risk of developing Type 2 DM, with maternal and fetal complications persisting as major concerns [50, 53]. IoMT solutions, including mobile health applications linked to CGMs and telemedicine services, improve monitoring and allow multidisciplinary teams to adjust therapy in real-time [49]. These approaches reduce hospital visits and enhance patient engagement, though barriers remain for women in low-resource or rural settings [48].

Table 4 presents major differences between the major two types of diabetes (Type-1 DM and Type-2 DM) [52] in terms of set of parameters such as age, Insulin resistance manifestations, etc.

4.2 Risk Factors and Complication of DM

There are many risk factors for developing diabetes that include genetic predisposition, obesity, lack of physical activity, stress and history of gestational DM. Genetic predisposition is linked to both type-1 and type-2 DM. while obesity, lack of physical activity, high fat diet and sugary beverage consumption, history of gestational DM and severe mental stress are linked to type-2 DM occurrence [54].

Improperly controlled diabetes is associated with many complications such as diabetic retinopathy, nephropathy, neuropathy, cardiovascular diseases and increased susceptibility to various infections [48]. Diabetic retinopathy is a serious complication that affect the diabetic patients' retina and can lead to eye damage and blindness. Diabetic nephropathy is a common diabetes complication than can

cause chronic renal damage and chronic renal failure [55]. Diabetic neuropathy also is a very common complication of diabetes which cause numbness and paresthesia interfering with patients' daily activity. Serious cardiovascular events can complicate DM including coronary atherosclerosis and acute myocardial infarction [56]. DM reduces the patients' immunity significantly to a level that increases their susceptibility to various infections and increases their complications [54].

4.3 Diagnosis and Management of DM

There are four criteria to diagnose patient with diabetes [50]: (a) Elevated fasting blood glucose level > 126 mg/dl, (b) Elevated two hours post prandial blood glucose > 200 mg/dl, (c) Elevated glycated hemoglobin (HBA1c) $> 6.5\%$, and (d) Random blood glucose > 200 mg/dl accompanied by symptoms of diabetes.

For diabetes management, there are several lines including (a) life style modifications, (b) oral therapy, (c) insulin injection and (d) other new treatment modalities [52]. Life style modifications include dietary changes, physical activity and controlling body weight. Dietary habits changes in the form of reducing fat and carbohydrate in diet and increasing vegetables and fibers [57]. Regular physical activity is essential for glycemic control as this physical activity increases insulin responsiveness of body cells. Body weight control in the form of weight loss in obese patients improves also glycemic control [58]. Insulin therapy is indicated to all type-1 diabetes patients and up to 30% of type-2 patients. In accurate insulin dose can result in either hyperglycemia or hypoglycaemia [54].

There are different oral drugs used for controlling blood glucose levels in diabetic patients. These drugs exert their action through different mechanisms and suit different conditions [54]. For example, α -glucosidase inhibitors decreases glucose absorption from gastrointestinal tract, biguanides decrease insulin resistance, insulin secretagogues enhances insulin secretion from β -cells, *GLP* – 1 agonists and *DPP4* inhibitors increases blood levels of insulin but the later carry risk of pancreatitis and *SGLT*-2 inhibitors perform their action through increasing urinary glucose excretion [59]. Each drug category can be used separately or in combination with other types [54, 60]. Recently, insertion of functioning Langerhans islets cells into patients' body thus the patient need no external insulin later on [54].

4.4 From Clinical Needs to System Requirements

The clinical properties of diabetes translate into four digital requirements for any technology stack:

- **R1** continuous or near-real-time sensing of glucose to detect impending hypo/hyperglycemia (CGM) [4, 6].
- **R2** timely remote support for alerts and dosing decisions (RPM/closed-loop assistance) [21, 22].
- **R3** trustworthy, tamper-evident storage and sharing of longitudinal health data [14, 25].
- **R4** privacy-preserving analytics for prediction and decision support [26].

These requirements provide a direct bridge from clinical risk to the technologies discussed next (see Fig. 1).

5 The Internet of Medical Things (IoMT)

IoT is defined as a network of STHs [16]. New paradigms are originated from the IoT to clarify its role in specific application, such Internet of Medical Things (IoMT), or its coupling with certain technology, such Chain of Things (CoT) and Internet of Artificial Things (IoAT) [24]. This paper presents Chain of Artificial Medical Things (CoAMT) as a new paradigm to integrate them. Addressing **R1 (continuous sensing)** and **R2 (remote support)**, the IoMT layer (wearables, sensors, and mHealth) enables CGM and RPM with continuous data capture and timely notifications (see Section 4.4).

5.1 IoMT Architecture

Authors in [61] and [62] study different architectures for the IoT and recommend the four-layers architecture for majority

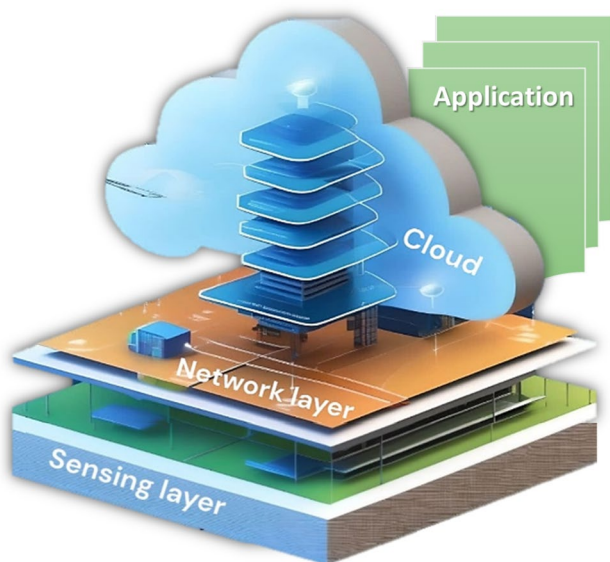


Fig. 6 Common IoT Architecture

of applications (Fig. 6). The following points summarize the main roles of these layers that from the IoMT perspective.

- **Sensing Layer:** serves as the initial point of contact with the physical environment for collecting data from the physical world through a network of sensors and actuators (i.e., STHs) [61, 63, 64], enabling the IoT ecosystem to acquire meaningful insights from the environment.
- **Network Layer:** plays a crucial role in establishing an efficient topology for forwarding data from the source node to the destination node within an IoT system [65].
- **Cloud Layer:** service layer in IoT relies on middleware technology, which provides functionalities to seamlessly integrate services and applications. This middleware plays a crucial role in facilitating smooth communication and interaction between different components within the IoT ecosystem [63].
- **Application Layer:** plays a crucial role, IoMT applications typically demand user-friendly interfaces for normal users. This layer should enhance API services to enable automatic calling of machine services.

5.2 IoMT Benefits, Applications, and Devices

IoMT is rapidly growing, offering significant advancements in patient care through real-time data monitoring and system integration [66]. According to Counterpoint [67], IoMT comes in the 3rd place in the top-10 IoT applications. Main benefits of IoMT applications are: (a) optimizing costs, (b) remote patient monitoring and control, (c) enriching medical records (tracking patient's vital signs), (d) enabling accurate real-time decisions (quick response), (e) improving presented medical services. Fig. 7 presents top-10 IoMT applications [68].

IoMT devices are categorized into six major categories: (a) wearable devices (e.g., smartwatches), (b) implantable devices (e.g., cardiac pacemaker), (c) stationary devices (e.g., smart beds), (d) remote monitoring tools (e.g., CGM), (e) hospital operation tools (e.g., hand hygiene), and (f) research and development tools (e.g., organ-on-a-chip) [69]. For diabetes prediction and management, wearable devices with mobile applications are notable developments [70, 71].

While IoMT technologies provide a wide range of benefits for diabetes care, they are also associated with certain challenges that need to be addressed for successful implementation. Table 5 summarizes the major pros and cons of IoMT in the context of diabetes management.

5.3 IoMT Challenges

Several significant challenges hindering the widespread adoption and efficient implementation of IoT technologies

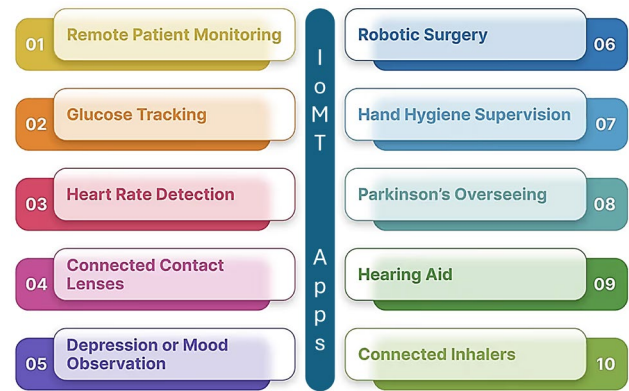


Fig. 7 Top-10 IoMT Applications [68]

Table 5 Pros and cons of IoMT in diabetes management

Pros of IoMT in Diabetes Management	Cons of IoMT in Diabetes Management
Continuous monitoring and early detection: Continuous glucose monitoring (CGM) systems and connected insulin pumps enable real-time glycemic tracking, which helps in reducing hypoglycemia and improving outcomes [49, 50].	Cost and accessibility: Diabetes care remains financially burdensome in many regions, as the high costs of insulin and modern devices limit accessibility, especially in developing countries [48, 50].
Personalized care: IoMT supports personalized medicine by integrating patient-specific data with AI/ML tools, allowing tailored therapy plans and optimized drug choices [49, 72].	Digital inequality: Low-resource settings face limited access to healthcare technologies, and disparities in digital literacy hinder adoption of IoMT solutions [16, 48].
Improved adherence: Wearables, smartphone apps, and telehealth systems improve treatment adherence and support lifestyle modification in diabetes patients, including children and adolescents [51, 52].	Data privacy and security: IoMT systems are highly vulnerable to cyberattacks and unauthorized data access, raising ethical and regulatory concerns [48, 61].
Remote consultations and reduced hospital visits: Telemedicine linked with IoMT devices provides continuous monitoring and remote support, which is particularly useful in gestational diabetes management [49, 53].	Over-reliance on technology: Increasing dependence on automated IoMT tools may lead to reduced patient self-management skills and clinical judgment errors [50].
Predictive analytics and prevention: Machine learning applied to IoMT datasets can predict complications and progression, enabling preventive interventions [49, 72].	Interoperability challenges: IoMT devices often lack standardization, creating integration issues with healthcare systems and reducing efficiency [16, 61].

Table 6 Layered Blockchain Architecture: Key Issues and Representative Solutions

Layer	Key issues	Representative solutions
Data	Ledger/storage growth; limited block capacity; transparency-driven privacy leakage and linkability [89, 90].	Efficient verification via Merkle trees and partial/lightweight clients; privacy-enhancing approaches (mixing, zero-knowledge-based schemes) [89, 90].
Network	P2P propagation delays; transient forks when blocks are found near-simultaneously; network-level isolation (eclipse-style) risks [89, 90].	Canonical chain selection to converge after forks (e.g., longest/heaviest subtree); robust peer connectivity and validation during sync [89, 90].
Consensus	PoW energy cost and low throughput; strategic behaviors (e.g., selfish mining); PoS-specific risks such as nothing-at-stake [89, 90].	Alternative/permissioned consensus when appropriate (PoS, PBFT/Tendermint, PoET) to improve efficiency; improved chain-selection rules and protocol mitigations against strategic attacks [89, 90].
Incentive	Centralization tendencies (e.g., mining pools); misaligned validator behavior when penalties are weak [90].	Reward/fee designs that favor decentralization and liveness; explicit penalties/slashing in PoS/BFT-style protocols to deter dishonesty [89, 90].
Contract	On-chain transparency may expose sensitive state/logic; correctness and security of contract code [90].	Smart-contract programmability with privacy-aware designs (as surveyed, e.g., zero-knowledge-based approaches); rigorous validation and verification of contract code [88, 90].
Application	Integrating with domain systems (e.g., healthcare interoperability), governance, and performance fit; mapping real processes to decentralized workflows [89, 90].	Domain patterns (smart-contract-based data sharing; blockchain-IoT integration); selecting public/private/consortium blockchains to match trust, scalability, and compliance needs [89, 90].

across various domains. IoT challenges have been classified into three main classes [61]: devices, network, and data which are related to IoT layers (sensing, network and cloud layers). This subsection spots the light on four major challenges that could restrict progress in the IoMT [73]. They are as follows:

5.3.1 Heterogeneity

Heterogeneity of SThs poses significant challenges at various levels of IoT deployment, including device interoperability, network, and data management [61, 74]. Thus, ontology is recommended for enabling interoperability and services orchestration [62].

- **Devices:** Different SThs utilize different protocols and possess varying capabilities, which leads to interoperability issues [24, 75].
- **Networks:** The IoT encompasses a complex and heterogeneous network environment, integrating various network types (WSNs, WLAN, cellular networks, etc.) and communication technologies [16, 24, 76].
- **Data:** IoT data comes from multiple sources and in various formats, leading to issues of inconsistency, inaccuracy, and quality [61, 65].

5.3.2 Scalability

presents a pressing challenge in IoT, requiring flexible network models and effective data processing strategies to accommodate the expanding network size and data volume.

- **Network Size:** With the exponential growth of SThs, ensuring the scalability of the IoT infrastructure becomes crucial [65, 77].
- **Data Volume:** The enormous volume of data generated by SThs presents challenges for data storage, processing, and analysis [61, 65]. Traditional data processing techniques are often inadequate, and big data solutions are needed to handle this influx of information [78].

5.3.3 Security and Privacy

- **Data Security:** Ensuring secure communication, data encryption, and access control mechanisms are essential for protecting sensitive information and preventing unauthorized access [79].
- **Privacy Protection:** The ability to track and monitor devices and individuals through the IoT raises ethical considerations and privacy concerns [61, 63].

5.3.4 Energy Efficiency

- **Device Level:** Many SThs rely on limited power sources, such as batteries, making energy efficiency a critical design consideration [65].

- Network Level: Efficient routing protocols and data transmission strategies can help to minimize energy consumption at the network level [61, 65]. Balancing network performance with energy efficiency is crucial for optimizing the overall sustainability of IoT systems.

5.4 Mitigation Strategies for IoMT Challenges

IoMT faces challenges such as data security, scalability, and real-time processing limitations. The following strategies address these concerns [80, 81].

- Strengthening Data Security with Blockchain: By integrating blockchain technology, IoMT networks can ensure immutable data storage and prevent data breaches [80]. Decentralized access control mechanisms reduce the dependency on third-party cloud storage, ensuring that patient records remain protected from unauthorized modifications.
- Optimizing IoMT Scalability with Cloud-Edge Hybrid Models: The combination of cloud computing and edge computing enables efficient processing of large volumes of IoMT-generated data [82]. This reduces network congestion and improves scalability for real-time diabetes management.
- Enhancing ML Models for Better Decision Making: AI-driven models continue to evolve, improving the accuracy of diabetes prediction and personalized treatment plans [81]. The use of federated learning further mitigates data privacy concerns by processing patient data without direct sharing [80].

5.5 IoMT Advancements for Diabetes Management

Traditional IoMT frameworks rely on cloud-based data processing. Recent advances incorporate edge computing and blockchain to improve privacy, scalability, and real-time analysis as indicated in the following points.

- Secure Health Data Management: Blockchain technology ensures secure and decentralized storage of diabetes-related health records, minimizing the risk of cyberattacks and unauthorized access by providing tamper-proof data storage [80]. Unlike cloud storage, blockchain enhances data integrity and patient privacy.
- Real-Time Glucose Monitoring: Edge computing processes glucose data closer to the source (e.g., CGM devices and insulin pumps), reducing latency and response time for insulin dosage adjustments. This enables real-time decision making, ensuring immediate intervention for diabetic patients [80].

- Diabetes Progression: AI models, including deep learning and federated learning, analyze historical glucose levels, lifestyle data, and genetic markers to predict diabetes progression with high accuracy [82]. Federated learning further improves data privacy by training models across multiple healthcare providers without exposing sensitive patient information [80].

6 Synergies Between IoMT, Blockchain, and AI for Diabetes Management

Recently, recent researches have spotted the light on harnessing Information and Communication Technology (ICT) in numerous fields and applications, especially in healthcare [5]. ICT encompasses a wide range of technologies [24, 83] such as AI, fog computing, big data, cloud computing, and blockchain technologies to manage and communicate information to build secure digital transformation blocks in IoT applications [24, 61].

To satisfy **R3 (data integrity/access control)** and **R4 (privacy-preserving analytics)** at scale, we augment IoMT with blockchain for integrity, auditability, and controlled access, and with AI/ML for forecasting glucose excursions and decision support (see Section 4.4).

6.1 Blockchain

The blockchain operates as a decentralized and immutable ledger, structured into blocks, each containing transaction data. These blocks are cryptographically linked, ensuring security and transparency[84]. The core components of blockchain include:

- Consensus Mechanisms: These protocols validate transactions without requiring a central authority. Popular mechanisms include:
 1. Proof of Work (PoW), used in Bitcoin, where miners solve complex mathematical problems to validate transactions[84].
 2. Proof of Stake (PoS), where the validators are chosen based on the number of tokens they have and are willing to "stake" [85].
 3. Delegated Proof of Stake (DPoS), a more scalable alternative where token holders vote for a set of validators[86].
- Smart Contracts: Self-executing contracts with predefined conditions that automate processes in healthcare applications, such as secure data sharing and automated insurance claims [87].

6.1.1 Layered Architecture of Blockchain

Blockchain can be systematically understood through a layered architecture, where each layer contributes to decentralization, transparency, and reliability. This layered view highlights how different functionalities build on one another to form a secure and adaptable ecosystem [88–90].

1. Data Layer The data layer defines the fundamental structure of blockchain blocks. Each block consists of a header (including version, Merkle root hash, timestamp, nonce, and parent block hash) and a body of transactions. This design ensures immutability, traceability, and tamper-resistance of records [89, 90].

2. Network Layer The network layer is responsible for peer-to-peer (P2P) communication among distributed nodes. It manages block propagation, transaction broadcasting, and ledger synchronization, thereby maintaining consistency across the decentralized network [88, 89].

3. Consensus Layer This layer ensures that all participating nodes agree on the validity of transactions and the order of blocks. Consensus algorithms such as Proof of Work (PoW), Proof of Stake (PoS), and Practical Byzantine Fault Tolerance (PBFT) provide fault tolerance and protect the blockchain from malicious forks or double spending [88–90].

4. Incentive Layer The incentive layer introduces mechanisms to reward participants such as miners or validators for their computational efforts. Rewards in the form of tokens or transaction fees sustain the network and encourage honest participation [88].

5. Contract Layer This layer incorporates programmability into blockchain through smart contracts. These self-executing agreements, stored on-chain, are triggered automatically under predefined conditions, enabling decentralized applications such as healthcare data sharing and automated insurance claims [88, 90].

6. Application Layer The application layer interacts with end-users and supports decentralized applications (dApps). It translates the underlying blockchain infrastructure into practical services, including secure electronic health records, pharmaceutical supply chain management, and IoT-enabled healthcare systems [89, 90].

In summary, the layered architecture demonstrates how blockchain achieves decentralization and trust: the data

layer secures information, the network and consensus layers ensure consistency, the incentive layer sustains participation, the contract layer provides programmability, and the application layer delivers services. A concise layer-wise summary of issues and representative solutions is provided in Table 2.

Blockchain technology is recommended in the healthcare sector, particularly for secure data sharing, remote patient monitoring, and improved privacy [24, 79, 91]. The following points summarize benefits of integrating blockchain in IoMT.

- **Security and Privacy:** The inherent characteristics of the blockchain, such as immutability, decentralization, and cryptography, offer enhanced security and privacy for sensitive medical data [21]. This helps address concerns about data breaches and unauthorized access, which are prevalent in traditional healthcare systems [1].
- **Data Integrity and Transparency:** The tamper-proof nature of the blockchain ensures data integrity and provides a transparent audit trail for all transactions [25]. This is crucial to maintaining trust and accountability in healthcare data management.
- **Improved Data Sharing and Interoperability:** Blockchain facilitates secure and efficient data sharing among healthcare stakeholders, including patients, providers, and researchers [1]. This can improve care coordination and facilitate medical research.
- **Patient Empowerment:** Blockchain-based systems can give patients greater control over their medical data, allowing them to decide who can access it and for what purpose [1].
- **Remote Patient Monitoring (RPM):** Integrating blockchain with IoT devices enables secure and efficient RPM, allowing healthcare providers to monitor patients remotely and intervene proactively in case of emergencies [91].

Traditional blockchain systems suffer from scalability issues, due to the computational demands of proof-of-work consensus mechanisms. This can limit their applicability in large-scale healthcare systems [21]. The energy requirements of blockchain technology, especially proof-of-work, can be a concern for resource-constrained IoMT devices (wearable).

6.2 Artificial Intelligence

AI is revolutionizing healthcare by automating diagnostics, predictive modeling, and treatment optimization[92]. Within AI, the key subfields include:

Table 2 Quality checklist

ID	Question
C1	IoMT relevance: explicitly targets diabetes management (CGM/RPM/insulin/telemedicine/wearables).
C2	Tech contribution: integrates AI/ML (prediction/decision support) and/or Blockchain/Security (privacy, access control, immutability).
C3	Evidence: empirical evaluation, simulation, pilot, or comparative analysis.

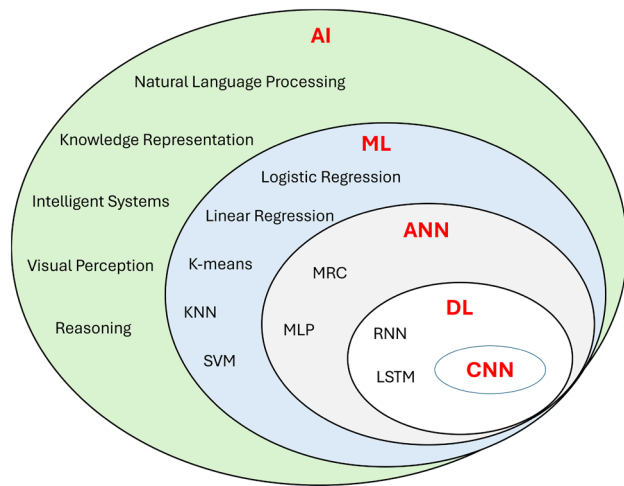


Fig. 8 ML, ANN, DL, and CNN as Common terminologies and their relationship to AI

- Machine Learning (ML): ML identifies patterns in diabetes-related data to improve disease prediction and treatment personalization[92].
- Deep Learning (DL): A subset of ML using neural networks to process large datasets, enhancing diagnostic accuracy in diabetes management [92].

Integrating AI into clinical workflows and EHR systems presents a significant challenge [93]. Data quality, bias, transparency, and explainability of AI algorithms are crucial concerns that need to be addressed [92]. Patient engagement and acceptance of AI-based solutions are also important factors for successful implementation [94]. The following subsections discuss the impact of the most common terminologies related to AI on diabetes: (a) Machine Learning (ML), (b) Deep Learning (DL), and (c) Convolutional Neural Network (CNN). Fig. 8 illustrates their hierarchical relationship within the broader field of AI [95–97]. In addition, Table 7 summarizes the most commonly used datasets in diabetes prediction research.

6.2.1 ML and Diabetes

Recent research has highlighted the potential of ML for diabetes prediction. Studies [108] and [109] have shown that ontology-based ML techniques, along with algorithms such as SVM, KNN, ANN, Naive Bayes, Logistic Regression, and Decision Tree, are effective in this context. Authors in [110] stressed the significance of dataset characteristics and model interpretability in determining performance.

6.2.2 DL and Diabetes

Authors in [111] and [112] emphasize the effectiveness of DL techniques in tasks such as diagnosing diabetes, managing glucose levels, and classifying outcomes. Authors in [113] further stresses the impact of DL in analyzing diabetic retinopathy, particularly in screening, segmenting images, and classifying results. In parallel, studies on DL's potential in diabetes research has been further demonstrated. For instance, authors in [111] and [114] highlights the efficacy of neural network algorithms in diagnosing and predicting diabetes, achieving enhanced accuracy.

6.2.3 CNN and Diabetes

Recent studies have demonstrated the effectiveness of CNNs in analyzing and detecting retinal conditions in diabetic patients, achieving 97% accuracy in classifying retinal images for diabetic retinopathy (DR) [115]. An improved CNN algorithm was used to predict changes in diabetic conditions, assisting in treatment planning and risk assessment [116]. Additionally, a CNN model with 18 convolutional layers and 3 fully connected layers was proposed for DR screening and staging, achieving high accuracy and reducing the dependency on specialized expertise [117]. These findings collectively underscore the potential of CNNs in improving the diagnosis and management of diabetes-related retinal disorders.

Table 7 Common Used Datasets and Their Characteristics

Ref.	Dataset	Features	no. of Features	Positive Classes	Negative Classes	Total Records
[47, 72, 98–102]	PIMA Indian	Gender, Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes pedigree	8	268 (34.9%)	500 (65.1%)	768
[47]	Sylhet	Age, Gender, Polyuria, Polydipsia, Sudden weight loss, Weakness, Polyphagia, Genital thrush, Visual blurring, Itching, Irritability, Delayed healing, Partial paresis, Muscle stiffness, Alopecia, Obesity	16	320 (61.5%)	200 (38.5%)	520
[47]	MIMIC III	Ethnicity, Gender, Age, Family History of Diabetes	4	N/A	N/A	46,520
[103, 104]	Diabetes 2019	Age, Gender, Family_Diabetes, highBP, PhysicallyActive, BMI, Smoking, Alcohol, Sleep, SoundSleep, RegularMedicine, JunkFood, Stress, BPLevel, Pregnancies, Pdiabetes, UrinationFreq	17	266 (28%)	684 (72%)	950
[105–107]	Diabetes Prediction	gender, age, hypertension, heart_disease, smoking_history, bmi, HbA1c_level, blood_glucose_level	8	8,500 (8.5%)	91,500 (91.5%)	100,000

6.3 Illustrative Pseudocode of Representative Pipelines

Note. The following listings are *illustrative pseudocode* synthesized from prior peer-reviewed works to reflect common workflows; they are not verbatim copies of any single implementation [118–121].

Require: IoMT device d , user u , permissioned ledger $\mathcal{L}$ (e.g., Hyperledger Indy), edge node e
Ensure: Authenticated session token T_{auth} and on-chain audit entry

```

1: (Enrollment) REGISTERDID( $u, \mathcal{L}$ ); ISSUECREDENTIAL( $u, d$ )
2: (Request)  $req \leftarrow \text{AUTHREQUEST}(d, u)$ 
3: (Verification) if VERIFYCREDENTIAL( $req, \mathcal{L}$ ) = true then
4:    $T_{\text{auth}} \leftarrow \text{GENERATE\_TOKEN}(u, e, \text{scope} = \text{glucose/telemet})$ 
5:   LOGEVENT( $\mathcal{L}, \text{deviceId} = d, \text{user} = u, \text{edge} = e, \text{time} = t$ )
6:   return  $T_{\text{auth}}$ 
7: else REJECTACCESS( $d$ ) end if

```

Algorithm 1 Hybrid Blockchain–Edge Authentication for IoMT Devices (based on [118, 121])

Require: Global model $\theta^{(t)}$, participants $\mathcal{P}$ (hospitals/devices), ledger $\mathcal{B}$
Ensure: Updated model $\theta^{(t+1)}$ and immutable training log

```

1: (Broadcast) BROADCASTMODEL( $\theta^{(t)} \rightarrow \mathcal{P}$ )
2: for all  $p \in \mathcal{P}$  in parallel do
3:    $\theta_p \leftarrow \text{LOCALTRAIN}(\theta^{(t)}, \text{data}_p)$ 
4:    $\Delta_p \leftarrow \theta_p - \theta^{(t)}$ 
5:   (Privacy Optional)  $\tilde{\Delta}_p \leftarrow \text{CLIPANDNOISE}(\Delta_p; S, \sigma)$ 
6:   (Integrity) COMMITUPDATE( $\mathcal{B}, H(\tilde{\Delta}_p), \text{id} = p$ )
7: end for
8: (Aggregation)  $\theta^{(t+1)} \leftarrow \text{AGGREGATE}(\{\tilde{\Delta}_p\}_{p \in \mathcal{P}})$   $\triangleright$  e.g., FedAvg
9: (Audit) LOGMODEL( $\mathcal{B}, H(\theta^{(t+1)}), \text{round} = t$ )
10: return  $\theta^{(t+1)}$ 

```

Algorithm 2 Federated Blockchain–IoMT Training Round for Healthcare Analytics (based on [119, 120])

Algorithm 1 mirrors permissioned blockchain+edge authentication and immutable audit for IoMT [118, 121]. Algorithm 2 follows standard FL training cycles with blockchain-backed integrity and logging for trustworthy model evolution [119, 120].

7 Industrial IoMT

Revolution of industrial hybrid closed-Loop products for automated Insulin delivery enable patients to smooth out glucose level control. Closed-loop systems are algorithm-driven based systems for automating insulin delivery [7]. There are several studies such as [7, 122, 123], and [124] for comparing commercial insulin delivery systems. A comparative study [4] recommends automated insulin delivery systems as a promising solution based on improved glucose control. Recently, [125] presented a comparative study in terms of the percentage of time of the three cases (a) time below range (TBR): $< 3.9 \text{ mmol/L}$, (b) time in range (TIR):

between $3.9 - 10 \text{ mmol/L}$, and (c) time above range (TAR): $> 10 \text{ mmol/L}$. They concluded that significant reductions in HbA1c result in Tandem Control-IQ (CIQ) and Omnipod 5 (OP5). Metallothionein (MT) and HbA1c level are of the main factors for optimizing time to deliver the insulin. Future research must focus on the main pillars for enabling automated insulin delivery: sensors, pumps, and algorithms [124].

Table 8 summarizes the industrial products dedicated to insulin delivery, as Device/System, Base Technology, and Description.

8 Future Directions and Recommendations

This section shed the light on required enhancements for enabling smart and secure diabetes management.

8.1 Future Directions for Improved Diabetes Management

Key services such as monitoring, diagnosis, and treatment approaches still are open issues for enabling truly diabetes management.

- **Monitoring:** real-time monitoring for vital signs on which other services like diagnosis rely on. Future research should focus on balanced monitoring for each use case (e.g., DM) [73], achieving optimal energy consumption, higher accuracy, secure transmission, balanced data volume, and truly personalized and continuous monitoring solutions [129].
- **Diagnosis:** the pursuit of accurate and painless blood glucose monitoring has spurred significant advancements in non-invasive techniques [130–132]. Early Detection using Novel Biomarkers is hot research point [133].
- **Insulin Delivery Approaches:** research in the field of diabetes auto-injection has focused on the development of intelligent insulin injection controllers to improve adherence to therapy [134, 135]. Table 8 indicates how Closed-Loop products are particularly important in the context of automated insulin delivery, which can help maintain ideal blood glucose levels [136].

8.2 Recommendations for Enabling Secure and Intelligent Diabetes Management Systems

CoAMT paradigm is proposed to build secure intelligent healthcare systems. Based on the previous studies and analysis work, for enabling features such as monitoring, detecting, and managing hypoglycemic attacks in diabetes case

Table 8 Industrial Hybrid Closed-Loop Products for Automated Insulin Delivery

Ref.	Year	Product	Base Technology	Description
[6, 7, 123, 126]	2023	Medtronic 670G/770G/780G	MiniMed 670G/770G/780G insulin pump Guardian Sensor 3/4 CGM SmartGuard algorithm in pump	First commercial hybrid closed-loop system approved by FDA in 2017 for ages +7. Available in US, Canada, some European/Middle Eastern countries. 670G automates basal insulin delivery every 5 mins based on CGM data and insulin history. 770G has improved pump with Bluetooth. MiniMed AID improves glycaemic control. Supports cloud storage. Upgraded version of 670G with Bluetooth, Remote updates, and Auto correction boluses. **Disadvantages** : No pregnancy license, requires multiple finger-sticks.
[7, 126]	2023	Tandem Control-IQ (CIQ)	Tandem t:slim X2 pump Dexcom G6 CGM Control-IQ algorithm in pump	CIQ algorithm predicts glucose 30 minutes ahead and adjusts insulin delivery. Allows higher glucose targets for exercise and nighttime. Requires setting a basal program for the algorithm. Approved by FDA for ages 14 and above. Controlled Enabled Pump (CE) marked for ages ≥ 6 years. Supports cloud storage. **Disadvantage** : No pregnancy license.
[7, 126]	2023	CamAPS FX	Android app Compatible with Dexcom G6, Libre 3 CGMs Works with DANA, DANA-i, Ypsomed pumps	Supports pregnancy license. CE marked for ages 1 and above. Available in select European countries. Allows multiple glucose targets. Licensed for use during pregnancy. Compatible with both rapid and ultra-rapid insulin. Requires a compatible Android smartphone. Supports cloud storage.
[126]	2023	Diabeloop DBLG1	Kaleido insulin pump Dexcom G6 CGM External command module with algorithm	Hybrid closed-loop system requiring meal announcements. Provides real-time CGM data viewing via web platform. CE marked for adults with T1D. Available in some European countries.
[126]	2023	Omnipod 5 (OP5) - HCL	Tubeless insulin pump Wireless handheld device Dexcom G6 CGM	Known as Hybrid Closed Loop (HCL). Consists of a tubeless insulin pump, a wireless handheld device, and the Dexcom G6 CGM. Delivers insulin micro-boluses every 5 minutes. Requires users to deliver mealtime boluses. Received FDA approval and CE mark for use in type 1 diabetes for individuals aged 2 years and older.
[127]	2022	Tidepool Loop (Open Source)	Open-source iOS app MPC algorithm Interoperable with various pumps and CGMs	Loop is an open-source iOS app developed by the DIY community. Uses an MPC algorithm to continuously modulate insulin delivery based on glucose levels. Interoperable with any alternate controller-enabled pump and CGM. 6-month study of 558 children and adults showed: 73% time in range and HbA1c of 6.5%. Used with Medtronic/Insulet pumps and Dexcom/Medtronic CGMs.
[126, 128]	2023	iLet Bionic Pancreas (Insulin Only)	MPC algorithm iLet insulin pump Dexcom G6 CGM	MPC algorithm, iLet insulin pump, Dexcom G6 CGM. Adapts in closed-loop mode. Automatically adjusts insulin doses. Adjustable from 6.1 to 7.3 mmol/L. Does not require pre-programmed basal rates when not in closed-loop mode. No randomized controlled trial data yet for insulin-only version.

study. CoAMT has to provide the following components and requirements:

- **User Registration and Authentication:**

- User registration for patients and doctors

- Secure authentication mechanisms (biometrics, two-factor authentication, etc.)

- **Data Collection and Management**

- IoT sensors and devices collect health information (continuous glucose monitoring sensor data)

- Data transmission protocols (Bluetooth, Wi-Fi, cellular networks, etc.)
- Data preprocessing and normalization
- **Fog Computing**
 - Edge computing devices process health data before forwarding it to the blockchain.
 - Data filtering, aggregation, and compression
 - Local decision-making and response mechanisms
- **Cloud Server and Machine Learning Model**
 - Cloud server hosts the machine learning model.
 - Model training and updating mechanisms.
 - Integrating blockchain for data retrieval and storage
- **Blockchain Implementation**
 - Secure storage of health-related data on the blockchain.
 - Data encryption using SHA256 hash method or other secure algorithms.
 - Suitable blockchain platform (e.g., Ethereum).
 - Access control mechanisms and data privacy measures.
- **Hypoglycemic Attack Detection and Management**
 - Machine learning model detects hypoglycemic attacks based on patient health data.
 - Alert and notification systems for patients, doctors, and authorized individuals.
 - Integration with auto-injection devices for insulin administration.
- **Auto-Injection of Insulin**
 - Continuous subcutaneous insulin infusion (CSII) pumps attached to patients.
 - Mobile application connected to the CSII pump for remote control and monitoring.
 - Automatic insulin injection based on detection results and prescribed dosages.
 - Real-time monitoring and reporting of insulin administration.
- **User Notifications and Information Sharing**
 - Detailed information about hypoglycemic attacks, health status, and treatment.
 - Notifications to patients, doctors, and authorized individuals (e.g., family members).
 - Secure sharing for EHRs.
- **Security and Privacy**
 - End-to-end encryption mechanisms
 - Secure communication protocols.
 - Access control mechanisms.
 - Role-based permissions.
 - Data privacy regulations (HIPAA, GDPR).
- **System Monitoring and Maintenance**
 - Enabling CGM and logging of system performance.
 - Fault tolerance and failover mechanisms
 - Software updates and security patches.
 - Scalability and load balancing mechanisms.

9 Conclusion and Future Work

This paper comprehensively explored the challenges and advancements in DM management, emphasizing how IoMT, blockchain, and AI can transform diagnosis, treatment, and long-term care. It starts with an overview of DM's types, complications, and management strategies, the study delved into how IoMT facilitates real-time monitoring and personalized care through innovative devices such as CGM systems. Despite its benefits, IoMT implementation faces hurdles like data security, privacy, and integration challenges.

This study highlights that Blockchain is a robust solution to security and privacy issues to manage sensitive health data. AI and ML further enhance IoMT applications by enabling predictive analytics, personalized treatment recommendations, and real-time decision making, paving the way for proactive and efficient diabetes care. Based on analysis results of latest industrial applications of IoMT for diabetes, security, privacy, accurate prediction, balanced monitoring, real-time intervention, and feedback from service providers are of the most important features that researchers have to focus on the future.

In addition, it is essential to recognize that the long-term success of AI- and IoMT-based solutions depends heavily on patient engagement and acceptance. Factors such as usability, digital literacy, trust in AI-generated recommendations, and concerns about privacy significantly influence adoption. Future research should therefore prioritize patient-centered design, transparency, and co-creation approaches to ensure that technological innovations are not only effective but also embraced by the people living with diabetes.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval not applicable.

Competing interests The authors declare no competing interests.

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